演算法觀點的圖論

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Preface

這個筆記是為了" 演算法觀點下的圖論" 這本書寫的筆記，只是把我覺得寫不清楚的地方多寫一點補充。

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1 圖論緣起

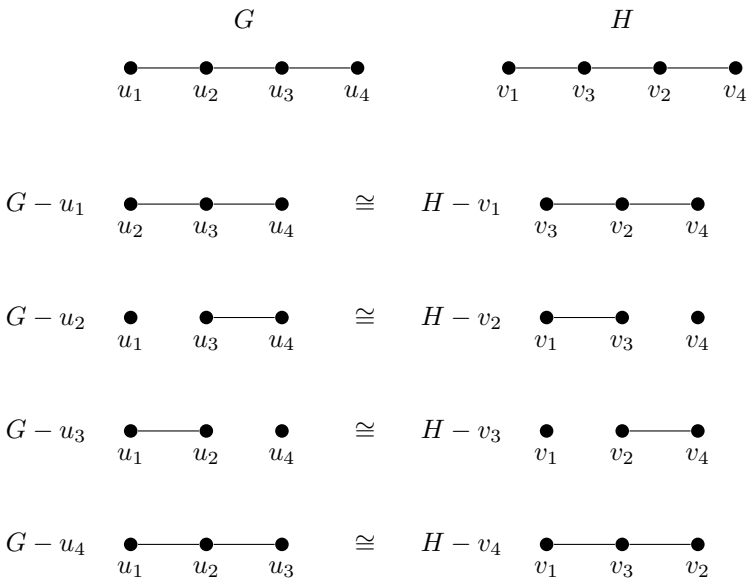
§1.1 話說 1736 年

§1.2 圖的定義

p.010

p.011

關於 Ulam 猜想的難點在於，如果有兩個圖：



雖然對於 $1 \leq i \leq 4$ 他們同構，但不是由點 u_i 對應到點 v_i 得到。

§1.3 路徑與連通

2 完美圖

§2.1 Shannon 零錯容量

p.260

這裡採用 Fekete's lemma 去證明 $\Psi(P) = \lim_{n\to\infty} N(n,P)^{\frac{1}{n}}$ 存在：

Lemma 2.1.1 (Fekete's lemma)

If $f : \mathbb{R} \to \mathbb{R}$ satisfies that $\forall m, n \in \mathbb{N}$, $f(m+n) \le f(m) + f(n)$, then

$$\lim_{n\to\infty} \frac{f(n)}{n}$$

exists and

$$\lim_{n\to\infty} \frac{f(n)}{n} = \inf_{n\ge 1} \frac{f(n)}{n}.$$

Proof. Let

$$\alpha \coloneqq \inf_{n\ge 1} \frac{f(n)}{n}.$$

Note that α must exist since $f(n) \ge 1$, and thus $\frac{f(n)}{n} \ge \frac{1}{n} > 0$, which shows $\left\{\frac{f(n)}{n}\right\}$ is lower-bounded.

Now since for every $n \in \mathbb{N}$ we have $\frac{f(n)}{n} \ge \alpha$, we have

$$\liminf_{n\to\infty} \frac{f(n)}{n} \ge \alpha.$$

It suffices to show that

$$\limsup_{n\to\infty} \frac{f(n)}{n} \le \alpha.$$

Let $\varepsilon > 0$, $\exists k \in \mathbb{N}$ s.t.

$$\frac{f(k)}{k} < \alpha + \varepsilon.$$

Now for every sufficiently large integer n, we write $n = qk + r$, where $0 \le r \le k-1$. Then, by sub-additivity,

$$\frac{f(n)}{n} \le q\frac{f(k)}{n} + \frac{f(r)}{n}.$$

We let

$$M \coloneqq \max_{0\le r\le k-1} f(r),$$

then

$$\frac{f(n)}{n} \le \frac{qf(k)}{n} + \frac{M}{n}.$$

Now $q = \frac{n-r}{k}$, we have

$$\frac{f(n)}{n} \le \frac{n-r}{k}\frac{f(k)}{n} + \frac{M}{n} = \frac{n-r}{n}\frac{f(k)}{k} + \frac{M}{n} \le \frac{f(k)}{k} + \frac{M}{n} < \alpha + \varepsilon + \frac{M}{n}.$$

Taking limsup on both sides, we have

$$\limsup_{n\to\infty} \frac{f(n)}{n} \leq \limsup_{n\to\infty} \alpha + \varepsilon + \frac{M}{n} = \alpha + \varepsilon.$$

Since this is true for all $\varepsilon > 0$, we have

$$\limsup_{n\to\infty} \frac{f(n)}{n} \leq \alpha,$$

and thus now we have

$$\alpha \leq \liminf_{n\to\infty} \frac{f(n)}{n} \leq \limsup_{n\to\infty} \frac{f(n)}{n} \leq \alpha.$$

This shows

$$\lim_{n\to\infty} \frac{f(n)}{n} = \alpha = \inf_{n\geq 1} \frac{f(n)}{n}.$$

■

Now we can show $\Psi(P) = \lim_{n\to\infty} N(n,P)^{\frac{1}{n}}$ exists by applying Fekete's lemma.

Proof. Suppose $f(n) = -\log N(n,p)$, then for any $m, n \in \mathbb{N}$ we have

$$\begin{aligned} f(m+n) &= -\log N(m+n,P) \leq -\log(N(m,P)N(n,P)) \\ &= -\log N(m,P) - \log N(n,P) = f(m) + f(n). \end{aligned}$$

Therefore, by Fekete's lemma,

$$\lim_{n\to\infty} \frac{f(n)}{n}$$

exists, which means

$$\lim_{n\to\infty} -\frac{1}{n} \log N(n,P)$$

exists, and since $f(x) = e^x$ is continuous,

$$\lim_{n\to\infty} N(n,p)^{\frac{1}{n}} = \lim_{n\to\infty} e^{\frac{1}{n}\log N(n,P)}$$

should also exist. ■